AR(2) in a Spatial Skew-t Model: Marginal Invariance, the Level-Autocovariance Ceiling, Predictive Scoring, and the Weak Identifiability of ϕ

Technical Note — `ar2_rethink`

Abstract

We document a connected sequence of derivations about the AR(2) temporal prior placed on the latent processes of a spatial skew-t model. The AR(2) construction normalises each latent so that its stationary marginal variance equals one; we prove this makes the time-t marginal of every latent invariant to the autoregressive coefficients ϕ. The Brier score under a spatial-only holdout integrates the predictive kernel against precisely this marginal, so the temporal coupling marginalises out: the AR(2) methods fail to dominate their i.i.d.-in-time siblings. We then sharpen the analysis in five directions. First, even a *perfectly specified* ϕ (frozen at the truth, zero estimation cost) cannot dominate, because the surviving posterior route delivers only an $O(q_t^{-2})$ variance reduction confined to non-spatial-noise components. Second — the central new result — we decompose the *level* autocovariance of the observed process exactly: $\mathrm{Cov}(Y_t, Y_{t+h}) = \lambda^2\gamma_z(h)$, so AR(2) reaches the level only through the skew channel λz; the AR(2) on the scale τ is pure stochastic volatility, invisible to every score that is a functional of the predictive law of the level. Because the simulation truth sets $\phi_z = (0, 0)$, the level of Y is *exactly white in time*, and no level-based score can reward the AR(2) prior for any holdout design — a temporal signal-to-noise ratio $\mathrm{SNR}_{\mathrm{temporal}}$ caps the achievable gain, with a closed-form CRPS ceiling $1 - \sqrt{1 - R(h)}$. Third, allowing ϕ to mix is more valuable for the convergence of the *other* parameters than pinning ϕ accurately. Fourth, a misspecified ϕ still reduces posterior variance but pays in bias; AR(2) collapses to i.i.d. only as $\phi \to 0$. Fifth, the observed collapse $\hat{\phi} \to 0$ is an errors-in-variables attenuation. We close with a scoring methodology — block-time holdout, per-horizon multivariate scores, marginal standardisation by the PIT, and a detectability calculus — under which the temporal structure that does exist becomes testable.

1 Introduction

- The simulation study in `code/analysis/simstudy/` compares eight prediction methods for spatial extremes; methods 7 and 8 place an AR(2) prior on every latent temporal process.
- Tabulated Brier scores for settings 9 and 10 show the AR(2) methods do *not* systematically beat their i.i.d. siblings; in setting 10 the $K = 5$ AR(2) variant is worse than the $K = 5$ no-AR baseline at every quantile.
- This note explains the discrepancy and the follow-on questions it raised. The conclusions concern the interaction of the AR(2) standardisation, the holdout geometry, the scoring rule, the MCMC sampler, and the identifiability of ϕ — not bugs in `simulate_ar2_standard()`.
- The decisive structural fact, isolated in Section 6, is that AR(2) enters the *level* of the observed field through a single channel. Two of the three AR(2) processes the model fits (τ and w) do

not move the level autocovariance at all, and the simulation truth switches the one that could (z) off.

The note is organised as follows. Section 2 fixes notation, the model, and a Gaussian surrogate used throughout. Section 3 proves the marginal invariance of the standardised AR(2). Section 4 carries that invariance through the Brier predictive integral and enumerates the leakage channels. Section 5 shows that even a perfectly specified ϕ cannot dominate. Section 6 derives the exact level-autocovariance decomposition, defines the temporal signal-to-noise ratio, and proves the CRPS ceiling. Section 7 analyses why a freely mixing ϕ helps the other parameters converge. Section 8 treats a misspecified ϕ and the robustness of the variance diagnostic. Section 9 derives the mechanism of the observed collapse $\hat{\phi} \to 0$. Section 10 develops the scoring methodology and Section 11 concludes.

2 Model Setup and Notation

2.1 Spatial skew-t knot model

Let $\{s_i\}_{i=1}^{n_s} \subset \mathbb{R}^2$ denote spatial locations and $t \in \{1, \ldots, T\}$ time indices. Fix K knots. The conditional model for the observed process is

$$Y_t(s) = x_t(s)^\top \beta + \lambda\, z_{g(s,t),t} + \frac{1}{\sqrt{\tau_{g(s,t),t}}} \big[L_C\, \varepsilon_t\big]_s, \qquad \varepsilon_t \sim \mathcal{N}(0, I_{n_s}), \tag{1}$$

where:

- $\beta \in \mathbb{R}^p$ are regression coefficients and λ is the skewness parameter;
- $z_{k,t}, \tau_{k,t}$ are knot-level latent processes governing skew and scale;
- $w_{k,t} \in \mathbb{R}^2$ is the (possibly time-varying) location of knot k at time t;
- $g(s,t) = \arg\min_k \|s - w_{k,t}\|$ is the knot membership function (a Voronoi assignment);
- $C = C(\rho, \nu, \gamma)$ is the spatial covariance matrix and $L_C = \mathrm{chol}(C)^\top$.

We collect the time-t slice of all latents into $\mathcal{L}_t = (z_{\cdot,t}, \tau_{\cdot,t}, w_{\cdot,t})$ and the fixed-parameter vector into θ. Throughout we abbreviate the conditional scale at a knot by

$$\sigma_{k,t} := \tau_{k,t}^{-1/2}, \tag{2}$$

so that the third term of (1) is $\sigma_{g(s,t),t}\,[L_C \varepsilon_t]_s$.

2.2 AR(2) standardisation

Each scalar latent process (one per knot per quantity z, τ, w) is generated by

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \eta_t, \qquad \eta_t \sim \mathcal{N}(0, \sigma_\eta^2), \tag{3}$$

with (ϕ_1, ϕ_2) in the stationarity region. We adopt the construction in `simulate_ar2_standard()`, which fixes the stationary variance and *solves* for the innovation variance.

Definition 1 (Standardised AR(2)). Set $\gamma_0 = 1$, $\gamma_1 = \phi_1/(1 - \phi_2)$, and

$$\sigma_\eta^2 = \gamma_0 - \phi_1\gamma_1 - \phi_2\gamma_2, \qquad \gamma_2 = \phi_1\gamma_1 + \phi_2\gamma_0. \tag{4}$$

The initial pair (X_1, X_2) is drawn jointly from $\mathcal{N}\big(0, \big[\begin{smallmatrix} 1 & \gamma_1 \\ \gamma_1 & 1 \end{smallmatrix}\big]\big)$, and the recursion (3) runs forward for $t \geq 3$.

Assumption 1 (Stationarity)**.** (ϕ_1, ϕ_2) satisfies $|\phi_2| < 1,\ \phi_1 + \phi_2 < 1,\ \phi_2 - \phi_1 < 1$, so the roots of the AR(2) characteristic polynomial lie strictly outside the unit circle.

2.3 Prediction task

The simstudy holdout fixes a subset $\mathcal{S}_{\text{test}}$ of test *sites*; the training set is $\mathcal{D} = \{Y_t(s) : s \notin \mathcal{S}_{\text{test}},\ t \in \{1, \ldots, T\}\}$, and the prediction targets are $\{Y_t(s^*) : s^* \in \mathcal{S}_{\text{test}}\}$. Every time point is observed at every training site.

Definition 2 (Brier score)**.** For threshold c and predictive probability $\hat{p}(s^*, t; c) = \mathbb{P}(Y_t(s^*) > c \,|\, \mathcal{D})$,

$$\mathrm{BS}(c) = \mathbb{E}\big[(\mathbf{1}\{Y_t(s^*) > c\} - \hat{p}(s^*, t; c))^2\big]. \tag{5}$$

2.4 Gaussian surrogate for one latent series

Sections 5–9 use a linear Gaussian surrogate that isolates a single knot's latent series.

Definition 3 (Gaussian surrogate)**.** Fix a knot k and a latent quantity; write the standardised AR(2) series as $X = (X_1, \ldots, X_T)^\top$. The prior is $X \sim \mathcal{N}(0, Q(\phi)^{-1})$, with $Q(\phi)$ the banded AR(2) precision matrix, normalised so $\operatorname{diag}(Q(\phi)^{-1}) = \mathbf{1}$ (Definition 1). After integrating the remaining parameters, the time-t training data act as a pseudo-observation

$$d_t = X_t + e_t, \qquad e_t \sim \mathcal{N}(0, q_t^{-1}), \quad \text{independent across } t, \tag{6}$$

with $D_q = \operatorname{diag}(q_1, \ldots, q_T)$. The posterior is $X \mid d \sim \mathcal{N}(\mu(\phi), \Sigma(\phi))$,

$$\Sigma(\phi) = \big(Q(\phi) + D_q\big)^{-1}, \qquad \mu(\phi) = \Sigma(\phi)\, D_q\, d. \tag{7}$$

We write $v_t(\phi) = [\Sigma(\phi)]_{tt}$ for the marginal posterior variance and $\kappa(\phi) = (1 + \phi_1^2 + \phi_2^2)/\sigma_\eta^2$ for the AR(2) full-conditional precision (an interior diagonal entry of $Q(\phi)$).

Remark 1. The pseudo-observations (6) are independent across t because the kernel (1) couples sites at a *common* t; the likelihood factorises over time given the latents. All cross-time structure resides in the prior $Q(\phi)$.

3 Marginal Invariance of the Standardised AR(2)

Proposition 1 (Marginal invariance to ϕ)**.** *Under Definition 1 and Assumption 1, for every t, $X_t \sim \mathcal{N}(0, 1)$ regardless of (ϕ_1, ϕ_2).*

Proof. By construction $X_1, X_2 \sim \mathcal{N}(0, 1)$ marginally. Suppose inductively that $\mathrm{Var}(X_{t-1}) = \mathrm{Var}(X_{t-2}) = 1$ and $\mathrm{Cov}(X_{t-1}, X_{t-2}) = \gamma_1$. Then from (3),

$$\mathrm{Var}(X_t) = \phi_1^2 + \phi_2^2 + 2\phi_1\phi_2\gamma_1 + \sigma_\eta^2.$$

Substituting (4) and $\gamma_2 = \phi_1\gamma_1 + \phi_2$ gives $\mathrm{Var}(X_t) = 1$. The mean is zero by linearity, and joint normality is preserved by the linear recursion. □

Corollary 1 (AR(1) toy)**.** *For an AR(1) process standardised to $\gamma_0 = 1$, $\sigma_\eta^2 = 1 - \phi^2$ and $X_t \sim \mathcal{N}(0, 1)$ for every admissible ϕ. The autocorrelation $\mathrm{Cov}(X_t, X_{t+h}) = \phi^{|h|}$ is the only quantity that moves with ϕ.*

Remark 2 (Identifiability)**.** ϕ is identifiable only from *joint* distributions across time; the one-point marginal is ϕ-blind. A single draw X_t carries zero information about ϕ. This remark is made quantitative in Section 9.

4 Why Brier Marginalises ϕ Away

4.1 The predictive integral collapses to the time-t marginal

Proposition 2 (Predictive at (s^*, t) uses only time-t latents)**.** *For the kernel* (1)*, the conditional CDF at (s^*, t) given $(\theta, \mathcal{L}_t)$ does not depend on $\mathcal{L}_{t'}$, $t' \neq t$. Hence*

$$\hat{p}(s^*, t; c) = \iint \mathbb{P}\big(Y_t(s^*) > c \,|\, \theta, \mathcal{L}_t\big)\, \pi(\mathcal{L}_t \,|\, \theta, \mathcal{D})\, \pi(\theta \,|\, \mathcal{D})\, d\mathcal{L}_t\, d\theta. \tag{8}$$

Proof. The right-hand side of (1) involves only $\mathcal{L}_t$ at the same t. Fubini and the factorisation $\pi(\theta, \mathcal{L}_t \,|\, \mathcal{D}) = \pi(\mathcal{L}_t \,|\, \theta, \mathcal{D})\, \pi(\theta \,|\, \mathcal{D})$ give the displayed form. □

Proposition 3 (ϕ-free prior marginal at one t)**.** *Under Definition 1, the* prior *marginal of $\mathcal{L}_t$ at any single t does not depend on $\phi := (\phi_z, \phi_\tau, \phi_w)$.*

Proof. Each component of $\mathcal{L}_t$ is a fixed monotone transform (gamma inverse copula for τ, half-normal inverse copula for z, inverse probit for w) of a standardised AR(2) latent. By Proposition 1 each latent has marginal $\mathcal{N}(0, 1)$ for every admissible ϕ, and the fixed transforms preserve this invariance. □

Corollary 2 (The posterior is the only route)**.** *Combining Propositions 2 and 3, ϕ affects $\hat{p}(s^*, t; c)$ only through the posterior $\pi(\theta \,|\, \mathcal{D})$ and the conditional $\pi(\mathcal{L}_t \,|\, \theta, \mathcal{D})$. The likelihood $p(\mathcal{D} \,|\, \theta)$ couples observations across time and so depends on ϕ; the kernel at the held-out cell does not.*

4.2 Four leakage channels

We enumerate the channels through which ϕ can still move the predictive at (s^*, t).

Channel A: $z_{g,t}$ via temporal pooling. AR(2) tightens $\pi(z_{k,t} \,|\, \mathcal{D})$ by borrowing from $z_{k,t\pm1}, z_{k,t\pm2}$. Bounded by the time-t likelihood precision.

Channel B: $\tau_{g,t}$ via temporal pooling. Symmetric to Channel A.

Channel C: knot membership $g(s^*, t)$. AR(2) tightens $\pi(w_{k,t} \,|\, \mathcal{D})$, but $g(s^*, t)$ is a discrete $\arg\min$; the tightening propagates only when a Voronoi boundary crosses s^*. For $K = 1$ the channel is closed.

Channel D: spatial noise $L_C \varepsilon_t / \sqrt{\tau_{g,t}}$. ε_t is i.i.d. in t given θ; ϕ does not appear in this channel at all.

Channel E (cost): posterior on ϕ. Methods 7 and 8 turn on AR(2) for z, τ, w simultaneously, introducing six parameters $\phi_{1z}, \phi_{2z}, \phi_{1\tau}, \phi_{2\tau}, \phi_{1w}, \phi_{2w}$ whose joint posterior adds variability to (8).

Lemma 1 (Predictive variance decomposition)**.** *For the held-out predictive probability $\hat{F}_{s^*,t}(c) = \hat{p}(s^*, t; c)$,*

$$\mathrm{Var}\big(\hat{F}_{s^*,t}(c)\big) \approx \mathbb{E}_\phi\big[\,\mathrm{Var}\big(\hat{F} \,|\, \phi\big)\big] + \mathrm{Var}_\phi\big(\mathbb{E}[\hat{F} \,|\, \phi]\big). \tag{9}$$

The AR(2) prior shrinks the first summand through Channels A–C and introduces the second (Channel E), which is identically zero under i.i.d. time. Brier is sensitive to the sum.

Table 1: Relative Brier score versus method 1 (Gaussian), setting 9. Entries below 1 indicate improvement over the baseline.

q	M2 (skew-t, K=1)	M4 (skew-t, K=5)	M7 (AR(2), K=1)	M8 (AR(2), K=5)
0.90	0.961	0.973	0.968	0.972
0.95	0.951	0.969	0.964	0.969
0.97	0.943	0.965	0.958	0.958
0.99	0.955	0.987	0.975	0.980
0.995	0.946	0.955	0.957	0.962

4.3 Evidence in the simstudy

The i.i.d. skew-t at $K = 1$ (method 2) systematically beats its AR(2) sibling (method 7). Setting 10 is sharper: method 8 ranges over 0.96–1.16 of the Gaussian baseline while the no-AR method 4 sits at 0.87–0.94.

5 A Perfectly Specified ϕ Still Cannot Dominate

We now ask whether a *frozen, true* ϕ — which eliminates Channel E entirely — would let AR(2) dominate. It does not.

5.1 Brier decomposition

Lemma 2 (Reducible/irreducible split)**.** *For a held-out cell, with $p(c) = \mathbb{P}(Y_t(s^*) > c)$ the true exceedance probability,*

$$\mathbb{E}[\mathrm{BS}(c)] = \underbrace{p(c)\,(1 - p(c))}_{irreducible} + \underbrace{\big(\mathbb{E}[\hat{p}] - p(c)\big)^2 + \mathrm{Var}(\hat{p})}_{reducible}. \tag{10}$$

The irreducible term is a property of the data-generating process and is ϕ-invariant; AR(2) can act only on the reducible term. With q_t training sites per knot informing each latent, the per-time likelihood swamps the prior, so both the AR(2) and i.i.d. models are unbiased to $O(q_t^{-1})$ and the contest reduces to $\mathrm{Var}(\hat{p})$.

5.2 The variance splits into a ϕ-reducible part and a ϕ-invariant floor

By the delta method, $\mathrm{Var}(\hat{p}) \approx [\partial_\mu \mathbb{P}(Y > c)]^2\ \mathrm{Var}(Y_t(s^*) \,|\, \mathcal{D})$, and

$$\mathrm{Var}\big(Y_t(s^*) \,|\, \mathcal{D}\big) = \underbrace{\lambda^2\, v_t^z}_{\text{Ch. A}} + \underbrace{(\cdots)\, v_t^\tau}_{\text{Ch. B}} + \underbrace{\mathbb{E}[\tau_{g,t}^{-1} \,|\, \mathcal{D}]\; C_{s^*s^*}}_{\text{Ch. D, } \phi\text{-invariant}}. \tag{11}$$

Channel D is the variance of the i.i.d.-in-time innovation $L_C \varepsilon_t$; it contains no ϕ and a perfect ϕ cannot shrink it.

5.3 The latent tightening is strictly positive but bounded

Proposition 4 ($O(q_t^{-2})$ bound on the frozen-ϕ benefit)**.** *In the surrogate of Definition 3 with constant $q_t \equiv q$, let $v_t^{\mathrm{iid}} = (1 + q)^{-1}$ be the i.i.d. posterior variance and $v_t(\phi)$ the AR(2) posterior* *variance. Then*

$$0 \;<\; \Delta v_t := v_t^{\mathrm{iid}} - v_t(\phi) \;\le\; \frac{\kappa(\phi)-1}{(1+q)\,(\kappa(\phi)+q)} \;=\; O\big(q^{-2}\big), \tag{12}$$

for every $\phi \neq 0$, where $\kappa(\phi) = (1+\phi_1^2+\phi_2^2)/\sigma_\eta^2 > 1$.

Proof. The i.i.d. prior contributes precision 1 (the $\gamma_0 = 1$ normalisation), so $v_t^{\mathrm{iid}} = (1+q)^{-1}$. The AR(2) posterior variance is bounded below by the case in which all neighbours are known exactly, which contributes the full-conditional precision $\kappa(\phi)$: $v_t(\phi) \ge (\kappa(\phi)+q)^{-1}$. Subtracting gives the upper bound. Strict positivity follows because $\sigma_\eta^2 \le 1$ forces $\kappa(\phi) \ge 1/\sigma_\eta^2 \ge 1$, with equality iff $\phi = 0$; any nonzero ϕ with an informed neighbour strictly tightens v_t. Expanding the bound for large q gives $\Delta v_t \lesssim (\kappa(\phi)-1)/q^2$. □

Corollary 3 (Frozen-ϕ verdict)**.** *A frozen true ϕ zeroes the second term of (9) but reduces the first only by $O(q^{-2})$, and only on the Channels A/B components of (11), never on the ϕ-invariant Channel-D floor. In the data-rich regime $q \gg 1$ the improvement is below the Monte Carlo resolution of a 50-dataset comparison: AR(2) fails to* dominate *not because ϕ is useless but because its benefit is bounded, partial, and crowded out.*

6 The Level-Autocovariance Ceiling

Section 5 bounded the *posterior-pooling* benefit of a frozen ϕ through a Gaussian surrogate. We now give an exact, model-level statement that does not rely on the surrogate: AR(2) reaches the *level* of $Y_t(s)$ through a single channel, and the simulation truth closes even that channel.

6.1 Exact decomposition of the level autocovariance

Fix a held-out site s^* with stable membership $g(s^*, t) \equiv g$ over the window of interest, and condition on the covariates so that $m_{s^*,t} := x_t(s^*)^\top \beta$ is deterministic. The stochastic part of the level is, using (2),

$$\widetilde{Y}_t := Y_t(s^*) - m_{s^*,t} = \lambda\, z_{g,t} \;+\; \sigma_{g,t}\, u_t, \qquad u_t := \big[L_C \varepsilon_t\big]_{s^*}, \tag{13}$$

where $\mathrm{Var}(u_t) = C_{s^*s^*}$, the u_t are i.i.d. across t, and $u \perp (z, \tau)$. The latent series z and τ are driven by separate AR(2) recursions and separate inverse-copula transforms, hence $z \perp \tau$.

Proposition 5 (Level autocovariance)**.** *Under model (1) with the independence structure above, for every lag $h \ge 1$*

$$\mathrm{Cov}\,\big(\widetilde{Y}_t,\, \widetilde{Y}_{t+h}\big) \;=\; \lambda^2\, \gamma_z(h), \qquad \gamma_z(h) := \mathrm{Cov}\,\big(z_{g,t},\, z_{g,t+h}\big). \tag{14}$$

Proof. Expand the product of (13) at times t and $t+h$ into four terms:

$$\mathrm{Cov}(\widetilde{Y}_t, \widetilde{Y}_{t+h}) = \lambda^2\, \mathrm{Cov}(z_{g,t}, z_{g,t+h}) + \lambda\ \mathrm{Cov}(z_{g,t}, \sigma_{g,t+h}u_{t+h}) + \lambda\ \mathrm{Cov}(\sigma_{g,t}u_t, z_{g,t+h}) + \mathrm{Cov}(\sigma_{g,t}u_t, \sigma_{g,t+h}u_{t+h}).$$

Both mixed terms vanish: u has mean zero and is independent of (z, σ), so e.g. $\mathbb{E}[z_{g,t}\sigma_{g,t+h}u_{t+h}] = \mathbb{E}[z_{g,t}\sigma_{g,t+h}]\,\mathbb{E}[u_{t+h}] = 0$, and the marginal factor $\mathbb{E}[\sigma_{g,t+h}u_{t+h}] = 0$ likewise. For the last term, $u \perp \sigma$ and the u_t are i.i.d. mean-zero across t, so for $h \ge 1$

$$\mathbb{E}[\sigma_{g,t}u_t\, \sigma_{g,t+h}u_{t+h}] = \mathbb{E}[\sigma_{g,t}\sigma_{g,t+h}]\ \mathbb{E}[u_t]\,\mathbb{E}[u_{t+h}] = 0,$$

and the product of marginal means is zero as well. Only the first term survives. □

Corollary 4 (The τ-AR(2) is pure stochastic volatility)**.** *The AR(2) on τ contributes* zero *to the level autocovariance* (14)*. Its temporal structure appears only in second-order moments: for $h \geq 1$,*

$$\mathrm{Cov}\left(\widetilde{Y}_t^{\,2},\, \widetilde{Y}_{t+h}^{\,2}\right) \;\supseteq\; C_{s^*s^*}^2 \;\mathrm{Cov}\left(\sigma_{g,t}^2,\, \sigma_{g,t+h}^2\right) \;\neq\; 0, \tag{15}$$

since $\sigma_{g,t}^2 = \tau_{g,t}^{-1}$ inherits the AR(2) dependence of τ. Hence τ-AR(2) is invisible to any scoring rule that is a functional of the predictive law of the level Y*, and visible only to rules sensitive to $|Y|^p$ (volatility clustering).*

Corollary 5 (The w-channel is second order)**.** *The AR(2) on knot locations w enters* (14) *only through the membership map $g(s^*, t)$. Under stable membership its contribution is zero; it becomes active only on the event that a Voronoi boundary sweeps across s^*, a second-order effect that is moreover absent for $K = 1$.*

Propositions 5 and its corollaries isolate the single level-channel: *the AR(2) prior moves the temporal autocovariance of the observable only through λz.*

6.2 The temporal signal-to-noise ratio and the CRPS ceiling

Definition 4 (Temporal signal-to-noise ratio)**.** With $\mathrm{Var}(\widetilde{Y}_t) = \lambda^2\, \mathrm{Var}(z) + \mathbb{E}[\sigma^2]\, C_{s^*s^*}$, define

$$\mathrm{SNR}_{\text{temporal}} \;:=\; \frac{\lambda^2\; \mathrm{Var}(z)}{\lambda^2\; \mathrm{Var}(z) + \mathbb{E}[\sigma^2]\, C_{s^*s^*}} \;\in\; [0, 1], \tag{16}$$

the share of level variance carried by the AR-bearing skew channel. The *forecastable fraction* at horizon h is

$$R(h) \;:=\; 1 - \frac{\mathrm{Var}(\widetilde{Y}_{t+h} \,|\, \text{past})}{\mathrm{Var}(\widetilde{Y}_{t+h})} \;=\; \mathrm{SNR}_{\text{temporal}} \cdot \kappa_z(h), \tag{17}$$

where $\kappa_z(h) := 1 - \mathrm{Var}(z_{t+h} \,|\, z_{\leq t}) / \, \mathrm{Var}(z) \in [0, 1]$ is the AR(2) forecastable variance fraction of z.

Proposition 6 (CRPS ceiling)**.** *Suppose the predictive law of the level is approximately Gaussian, so that for a calibrated forecast $\mathbb{E}\,\mathrm{CRPS} = \sigma_{\text{pred}}/\sqrt{\pi}$. Conditioning on the past reduces the predictive standard deviation from σ_Y (the marginal, no-AR model) to $\sigma_Y\sqrt{1 - R(h)}$. Hence the best-case relative improvement of the AR(2) model over the i.i.d.-in-time model is*

$$\frac{\Delta\,\mathrm{CRPS}}{\mathrm{CRPS}} \;=\; 1 - \sqrt{1 - R(h)}. \tag{18}$$

Proof. For $Y \sim \mathcal{N}(\mu, \sigma^2)$ the closed form of the CRPS gives $\mathbb{E}_Y \mathrm{CRPS}(\mathcal{N}(\mu, \sigma^2), Y) = \sigma/\sqrt{\pi}$, linear in σ. The marginal model predicts with standard deviation σ_Y; the AR(2) model, conditioning on the past, predicts with $\sigma_Y\sqrt{1 - R(h)}$ by (17). The ratio of expected scores is $\sqrt{1 - R(h)}$ and the relative improvement is its complement. □

Table 2: The CRPS ceiling (18) as a function of the forecastable fraction $R(h)$. The gain is concave and small for the R values plausible here.

$R(h)$	0.05	0.10	0.20	0.30
$1 - \sqrt{1 - R(h)}$	0.025	0.051	0.106	0.163

Remark 3 (There is no universal threshold)**.** A value such as $\mathrm{SNR}_{\text{temporal}} \approx 0.1$ is *not* a constant from the spatial-extremes literature; it is simply the regime in which the best-case relative CRPS gain at $h = 1$ is low single digits (Table 2), comparable to the Monte Carlo standard error of a few dozen holdout blocks. Whether a gain is detectable is the separate calculation of Section 10.7, not a comparison against a magic number. The Brier score is strictly weaker than CRPS here: it binarises at a single threshold $\{Y > c\}$ and so sees the temporal signal through one crossing only — a further information loss on top of the ceiling (18).

6.3 The simulation truth sets the ceiling to zero

Corollary 6 (Level whiteness under the experiment's truth)**.** *The data-generating process of settings 9–10 sets $\phi_z = (0, 0)$ (Eq. (30) below). Then z is i.i.d. in time, $\gamma_z(h) = 0$ and $\kappa_z(h) = 0$ for all $h \geq 1$, so by Proposition 5 and Definition 4,*

$$\mathrm{Cov}\left(\widetilde{Y}_t, \widetilde{Y}_{t+h}\right) = 0, \qquad R(h) = 0 \qquad \textit{for all } h \geq 1. \tag{19}$$

The level of Y is exactly white in time. *No level-based scoring rule — Brier, $\sum$CRPS, or the energy score on levels — can reward the AR(2) prior, for any holdout design and for any specification of ϕ (estimated, true-initialised, or frozen).*

This is the exact counterpart of Corollary 3: not merely *the benefit is small* but *the level-channel benefit is identically zero under the experiment's own truth.* The true temporal structure of settings 9–10 lives entirely in τ (volatility, Corollary 4) and w (membership drift, Corollary 5), and is reachable only by volatility- or dependence-sensitive scores (Section 10).

Remark 4 (Two confounds to control)**.** First, $\gamma_z(h)$ in (14) is the autocovariance of the *transformed* z (post inverse-copula); since the transform is a fixed monotone map of the standardised AR(2) latent $z^\star$, $\gamma_z(h)$ inherits the geometric decay of $z^\star$ but with a distorted magnitude — the diagnostic of Appendix C measures it directly rather than assuming it. Second, if the covariate $x_t(s^*)$ itself carries temporal autocorrelation, part of any observed $\mathrm{Cov}(Y_t, Y_{t+h})$ is covariate-driven and must be removed by conditioning on x before attributing a score difference to AR(2).

7 MCMC Mixing: ϕ as a Compensating Degree of Freedom

A separate experiment compared three sampler strategies for ϕ: (1) *update*; (2) *update from a true initialisation*; (3) *freeze at the true value.* The performance metrics were near-identical, but strategy 3 mixed *worst* despite knowing the truth (Table 3).

Table 3: Three ϕ strategies: 144 locations, 50 times, 5 knots; means over replicates (SD in parentheses). Lower MSPE and runtime are better.

Metric	Update	Update (true init)	Freeze (true)
MSPE	2.4953 (0.329)	2.4417 (0.344)	2.5427 (0.497)
Brier@0.95	0.0170 (0.003)	0.0169 (0.004)	0.0171 (0.004)
Time (s)	4652.6 (21.9)	4798.6 (21.2)	4810.2 (26.7)

7.1 Conditional stiffness versus marginal softness

Proposition 7 (Frozen ϕ targets a stiffer chain)**.** *Single-site updates of the latent field explore the conditional posterior $\pi(X \mid \phi, \mathcal{D})$; a sampler that also updates ϕ explores the marginal $\pi(X \mid \mathcal{D}) = \int \pi(X \mid \phi, \mathcal{D})\, \pi(\phi \mid \mathcal{D})\, d\phi$. The two are related by*

$$\mathrm{Var}(X \mid \mathcal{D}) = \mathbb{E}_\phi\big[\,\mathrm{Var}(X \mid \phi, \mathcal{D})\big] + \mathrm{Var}_\phi\big(\mathbb{E}[X \mid \phi, \mathcal{D}]\big), \tag{20}$$

so the marginal target is softer *(larger variance) in the smooth low-frequency directions than the conditional target at any fixed ϕ, including $\phi = \phi_{\mathrm{true}}$.*

The banded precision $Q(\phi)$ has small eigenvalues on its smooth eigenvectors; at fixed persistent ϕ the conditional posterior of X is strongly internally correlated along those modes, and single-site updates crawl. Freezing ϕ — even at the truth — locks the chain onto this stiff slice; updating ϕ targets the softer marginal of (20).

7.2 ϕ as a parameter-expansion coordinate

Remark 5 (Compensating degree of freedom)**.** The latent field's smoothness is a nuisance that τ, the knots, and β all interact with. When ϕ is frozen, the smoothness is a hard constraint and a τ/knot proposal that would prefer a smoother field is rejected; when ϕ is free, ϕ absorbs the coordinated adjustment and the proposal is accepted. This is the mechanism of parameter-expanded data augmentation [1]: an extra coordinate aligned with a stiff direction improves the conditioning of the sampler for every coupled block. The centered/non-centered and interweaving literature [2, 3] describes the same phenomenon.

Remark 6 (Two roles of ϕ)**.** ϕ has a role as an *estimand* — whose point value barely affects the held-out Brier (Sections 4–6) — and a role as a *mixing coordinate* — whose *freedom to move* conditions the sampler for τ, the knots, and z. Table 3 is explained by the second role: "frozen at truth" is not "free." Updating ϕ is also fastest in wall-clock because ϕ is low-dimensional, so the mixing gain is bought at negligible per-iteration cost.

8 A Misspecified ϕ: Robust Variance, Fragile Bias

We now treat the opposite extreme: ϕ estimated far from the truth.

8.1 The variance diagnostic is robust to a wrong ϕ

Proposition 8 (Posterior variance depends only on the assumed ϕ)**.** *In the surrogate of Definition 3, $v_t(\phi) = [\Sigma(\phi)]_{tt}$ depends on the* assumed *ϕ and D_q only — not on the true ϕ^*. Moreover*

$$v_t(\phi) = \mathrm{Var}(X_t \mid d_{1:T}) \;\le\; \mathrm{Var}(X_t \mid d_t) = \frac{1}{1+q_t} = v_t^{\mathrm{iid}} \tag{21}$$

for every stationary ϕ, with strict inequality whenever $\phi \neq 0$ and some neighbour carries data.

Proof. $v_t(\phi)$ is a function of $Q(\phi) + D_q$ only. Under the AR prior X_t is correlated with $d_{t'}$, $t' \neq t$, so conditioning on those extra pseudo-observations cannot increase the variance; under the i.i.d. prior $X_t \perp d_{t'}$ and they are inert, giving $v_t^{\mathrm{iid}} = (1+q_t)^{-1}$. □

Corollary 7 (The SD plot certifies coupling, not correctness)**.** *A per-cell posterior-SD plot of $X_{k,t}$ shows $\sqrt{v_t(\phi)} < \sqrt{v_t^{\mathrm{iid}}}$ for* any *nonzero ϕ, with the largest relative reduction at data-poor cells (small q_t). The plot is a diagnostic that the coupling is* active*, not that ϕ is* correct*.*

8.2 Bias is where a wrong ϕ is punished

Using $\Sigma(\phi)D_q = I - \Sigma(\phi)Q(\phi)$, the posterior mean has conditional bias

$$\mathbb{E}[\mu(\phi) \mid X] - X = -\,\Sigma(\phi)\,Q(\phi)\,X, \tag{22}$$

so a smoother built on the wrong ϕ shrinks X along the wrong correlation directions. The reconstruction risk

$$R_t(\phi;\phi^*) = \mathbb{E}\big[(\mu_t(\phi) - X_t)^2\big] = R_t(\phi^*;\phi^*) + \tfrac{1}{2}(\phi-\phi^*)^\top H_t(\phi-\phi^*) + o\big(\|\phi-\phi^*\|^2\big) \tag{23}$$

is minimised at $\phi = \phi^*$. Small misspecification costs only second order (robust); a large or wrong-sign ϕ can push $R_t(\phi;\phi^*)$ above the i.i.d. risk even while, by Proposition 8, the SD plot still looks tighter.

8.3 Collapse to i.i.d. occurs only as $\phi \to 0$

Proposition 9 (Collapse condition)**.** *As $\phi \to 0$, $Q(\phi) \to I$, $\Sigma(\phi) \to (I+D_q)^{-1}$ (diagonal), $\mu(\phi) \to$ the i.i.d. posterior mean, and $v_t(\phi) \to v_t^{\mathrm{iid}}$. AR(2) degenerates to i.i.d.* continuously and only *in this limit; a merely wrong but nonzero ϕ keeps the coupling fully active.*

Remark 7 (Weak identifiability resurrects Channel E)**.** If $\pi(\phi \mid \mathcal{D})$ is wide, the predictive averages smoothers:

$$v_t^{\mathrm{BMA}} = \underbrace{\mathbb{E}_{\phi \mid \mathcal{D}}[v_t(\phi)]}_{\le v_t^{\mathrm{iid}}} + \underbrace{\mathrm{Var}_{\phi \mid \mathcal{D}}(\mu_t(\phi))}_{\text{Channel E, reborn}}. \tag{24}$$

Weak identifiability either collapses to i.i.d. (if $\pi(\phi)$ pulls to 0) or keeps the coupling on while reinflating the estimation-variance cost.

9 The Mechanism of the Observed Collapse $\hat{\phi} \to 0$

The empirical posteriors show Case 1 of Remark 7: $\hat{\phi}$ shrinks aggressively to 0 even though the data-generating process carries a true temporal signal, $\phi_\tau = (0.8, -0.6)$. We derive why.

9.1 ϕ-information is monotone increasing in q_t

Proposition 10 (Marginal likelihood of ϕ)**.** *In the surrogate of Definition 3, the marginal likelihood of ϕ from one knot series is*

$$2\log L(\phi) = \log|Q(\phi)| - \log|Q(\phi)+D_q| + d^\top D_q (Q(\phi)+D_q)^{-1} D_q\, d + \text{const.} \tag{25}$$

Its limiting behaviour is

$$D_q \to 0: \quad 2\log L(\phi) \to \text{const} \quad (\phi \textit{ unidentified}); \tag{26}$$

$$D_q \to \infty: \quad 2\log L(\phi) \to \log|Q(\phi)| - X^\top Q(\phi)\,X \quad (\textit{complete-data AR(2) likelihood}). \tag{27}$$

The Fisher information $\mathcal{I}(\phi)$ interpolates monotonically between the two: richer per-time data increases *the information about ϕ.*

Proof. Equation (25) is the standard Gaussian integral $\int \exp(-\frac{1}{2}(X-d)^\top D_q(X-d) - \frac{1}{2}X^\top Q(\phi)X)\,dX$. For $D_q \to 0$ the determinant terms cancel and the quadratic form vanishes. For $D_q \to \infty$, $(Q+D_q)^{-1} \to D_q^{-1} - D_q^{-1}QD_q^{-1}$ and $d \to X$, leaving the complete-data log-likelihood. $\square$

The hypothesis "rich q_t masks the temporal linkage" is thus inverted: rich q_t is the only thing that reveals ϕ. A collapse $\hat{\phi} \to 0$ implies the *effective* q_t for the relevant latent is small.

9.2 Errors-in-variables attenuation

Lemma 3 (Attenuation of the visible autocorrelation)**.** *The pseudo-data series of Definition 3 has covariance* $\mathrm{Cov}(d) = Q(\phi^*)^{-1} + D_q^{-1}$*, i.e. AR(2)* plus white noise*. With constant* $q_t \equiv q$*, its lag-h autocorrelation is*

$$\rho_h^{\text{visible}} = \frac{\gamma_h}{\gamma_0 + q^{-1}} = \underbrace{\frac{q}{q+1}}_{\textit{attenuation } a} \rho_h^{\text{true}}, \qquad h \geq 1. \tag{28}$$

Fitting the pure AR(2) model recovers ϕ *from the attenuated autocorrelations; as* $a \to 0$*,* $\hat{\phi} \to 0$*.*

For $\phi_\tau = (0.8, -0.6)$ the true lag-1 autocorrelation is $\gamma_1 = \phi_{1\tau}/(1-\phi_{2\tau}) = 0.8/1.6 = 0.5$. With K knot series of length T, $\mathrm{SE}(\hat{\rho}_1) \approx (KT)^{-1/2}$, so ϕ is statistically alive only if

$$\frac{q}{q+1}\gamma_1 \gtrsim \frac{1}{\sqrt{KT}} \quad \Longrightarrow \quad q \gtrsim 0.15 \qquad (\gamma_1 = 0.5,\ K=5,\ T=50). \tag{29}$$

An aggressive collapse therefore implies an effective q_t^τ of order 0.1–0.3.

9.3 Why q_t^τ is structurally small

The raw site count is not the effective precision for a latent *scale*. Six losses compound:

1. scale parameters carry less Fisher information than means;
2. sites in a knot cell are spatially correlated through $C(\rho, \nu, \gamma)$, so the effective sample size is far below the cell count;
3. residual dispersion is jointly explained by τ, C, ν, and z — the information is split;
4. τ is the mixing variable of the t-distribution, weakly identified per draw;
5. the gamma inverse-copula layer between τ and $\tau^\star$ loses further information;
6. knot membership $g(s,t)$ drifts in t, so the informing site set is unstable.

9.4 The Gibbs feedback turns attenuation into collapse

Remark 8 (Fixed point of the Gibbs map)**.** The two full conditionals form a self-reinforcing map:

$$\phi \mid X \;\longrightarrow\; \text{small if } X \text{ looks rough}, \qquad X \mid \phi \;\longrightarrow\; \text{rough if } \phi \text{ is small}.$$

If $\phi \approx 0$, the smoother is i.i.d., each X_t is pulled only by its noisy d_t, the realised series has attenuated autocorrelation $a\,\gamma_1$, and $\pi(\phi \mid X)$ concentrates near 0 again. $\phi \approx 0$ is a stable fixed point; a prior centred at or diffuse around 0 selects its basin. That a true-initialised chain also collapses indicates the *marginal* posterior — not merely the basin — favours small ϕ.

9.5 The data-generating process is adversarial

The truth places the temporal signal in exactly the latents with the weakest per-time information:

$$\phi_z = (0, 0), \qquad \phi_\tau = (0.8, -0.6), \qquad \phi_w = (0.9, 0). \tag{30}$$

The latent that *would* be well identified is z — a location-type shift with large q_t^z — and its true ϕ is exactly zero. The signal lives in τ (a scale, tiny q_t^τ) and w (a knot location read through a discrete $\arg\min$, tiny q_t^w). Combined with Corollary 6, this is doubly adversarial: the one latent whose AR(2) would have reached the *level* of Y is precisely the one the truth switches off.

10 Scoring Redesign and Evaluation Methodology

The diagnosis above dictates the remedy. The level of Y is white in time (Corollary 6), so a level-based score on a spatial holdout cannot reward AR(2). Three things must change: the holdout geometry, the scoring rule, and the marginal scale on which it acts.

10.1 Averaged CRPS cannot see dependence

Proposition 11 (Averaged CRPS is proper for marginals only)**.** *Let* $\overline{\mathrm{CRPS}}(F) = \frac{1}{n_s T}\sum_{s,t} \mathrm{CRPS}(F_{s,t}, y_{s,t})$, *where* $F_{s,t}$ *is the predictive marginal at cell* (s,t)*. Then* $\overline{\mathrm{CRPS}}$ *is strictly proper for the collection of marginals* $\{F_{s,t}\}$ *but* not *strictly proper for the joint law* F*: any two joint forecasts with identical marginals receive identical expected* $\overline{\mathrm{CRPS}}$.

Proof. $\mathrm{CRPS}(F_{s,t}, \cdot)$ is a functional of the marginal $F_{s,t}$ alone. Its expectation under the true joint law depends only on the true marginals; hence every joint forecast whose marginals match the truth attains the same minimal expected score, irrespective of its copula. □

Consequently $\overline{\mathrm{CRPS}}$ is the correct *marginal-calibration control* but is blind to the cross-time and cross-site dependence that AR(2) is meant to supply. The dependence test must come from a genuinely multivariate score (Section 10.4).

10.2 Design A: block-time-forecast holdout

Hold out a *terminal* block $\mathcal{B} = \{T - T_h + 1, \ldots, T\}$ and train on $\{Y_t(s) : t \le T - T_h\}$. The predictive at horizon $t^* \in \mathcal{B}$ requires the AR(2) conditional

$$z_{k,t^*} \mid \mathcal{D} \sim \mathcal{N}\big(\phi_1 \hat{z}_{k,t^*-1} + \phi_2 \hat{z}_{k,t^*-2},\ \sigma_\eta^2 + \text{filter uncertainty}\big), \tag{31}$$

which is strictly ϕ-dependent — unlike the spatial holdout, the temporal channel is now load-bearing.

Remark 9 (Implementing the predictive loop)**.** Two implementations target different distributions. Treating $\mathcal{B}$ as missing data inside the Gibbs sampler samples $p(y_\mathcal{B} \mid y_\mathcal{O}) = \int p(y_\mathcal{B} \mid \theta, \mathcal{L})\, p(\theta, \mathcal{L} \mid y_\mathcal{O})\, d(\theta, \mathcal{L})$. For a *terminal* block this integral *is* the forecast distribution and in-sampler imputation coincides with a correct forward simulation. For an *interior* block it conditions on future observations and yields a *smoothing* distribution, which is not a forecast and flatters the AR(2) model unfairly. Recommendation: hold out a terminal block and use the model's missing-data imputation inside Gibbs — it propagates $(\theta, \mathcal{L})$ uncertainty jointly. A post-hoc forward simulation is also valid but must, per posterior draw, run the AR(2) recursions of `rpotspatTS_arp()` forward from the drawn terminal latent states; substituting posterior means collapses $\mathrm{Var}(y_\mathcal{B})$ and corrupts every score.

10.3 Per-horizon scoring and effective memory

AR(2) forecast skill decays geometrically. Let $\rho^* = \max_i |\lambda_i|$ be the dominant modulus of the AR(2) companion matrix; then the forecastable variance fraction obeys

$$\kappa_z(h) \asymp (\rho^*)^{2h}, \qquad h^* := \frac{\log(0.05)}{2\log\rho^*}, \tag{32}$$

with h^* the *effective memory* (where κ_z falls to 0.05): $\rho^* = 0.6 \Rightarrow h^* \approx 3$, $\rho^* = 0.8 \Rightarrow h^* \approx 7$. Two consequences:

- Choose the block length $T_h \approx$ 2–3 h^* — long enough to see the decay, short enough that informative horizons are not diluted by null ones.
- *Score per horizon*, reporting $S(h)$ as a curve in h rather than a single pooled number; pooling $T_h = 50$ averages ~ 5 informative horizons with ~ 45 null ones. Overlay the ceiling $1 - \sqrt{1 - R(h)}$ from Proposition 6 as the theoretical bound.

10.4 Multivariate scores: energy and variogram

Score the held-out block as a single vector $\boldsymbol{y} \in \mathbb{R}^d$, $d = n_s \cdot T_h$. The energy score [4]

$$\mathrm{ES}(F, \boldsymbol{y}) = \mathbb{E}_F\|X - \boldsymbol{y}\|_2 - \tfrac{1}{2}\mathbb{E}_F\|X - X'\|_2, \qquad X, X' \overset{\text{iid}}{\sim} F, \tag{33}$$

is the multivariate generalisation of the CRPS (CRPS is the $d = 1$ case) and is consistent, but has low power to detect misspecified dependence once the marginals are correct. The variogram score of order p [5]

$$\mathrm{VS}_p(F, \boldsymbol{y}) = \sum_{i<j} w_{ij}\big(|y_i - y_j|^p - \mathbb{E}_F|X_i - X_j|^p\big)^2 \tag{34}$$

is built from pairwise differences and is the sensitive dependence diagnostic. Recommendations, following [5, 6]:

- Use $p = 0.5$: the choice $p = 2$ is dominated by the largest pairs (a few extreme cells swamp the signal), whereas $p = 0.5$ is robust under heavy tails.
- Weight w_{ij} by inverse space–time separation, e.g. $w_{ij} = (1 + \alpha\|s_i - s_j\|^2 + \beta(t_i - t_j)^2)^{-1}$, to downweight uninformative far pairs.
- Compute via the unbiased U-statistic estimators in the `scoringRules` package (`vs_sample`, `es_sample`, `crps_sample`) [6] rather than a naive plug-in, which is biased.
- For an extremes-focused question, weight the score toward the tail with a threshold-weighted kernel [8]; condition with care, since naïvely subsetting on the realised event breaks propriety [7, 9].

10.5 Marginal standardisation: the PIT transform

Both (33) and (34) are sensitive to marginal scale; without standardisation a few high-variance sites dominate and the score conflates marginal fit with dependence fit. Apply the probability integral transform (PIT) $u_{s,t} = \hat{F}_{s,t}(y_{s,t})$ cell-by-cell before scoring.

- **Uniform** margins isolate dependence; recommended as the primary scale, since the scientific question is whether AR(2) improves the joint structure.

- **Gumbel** margins retain comparable magnitude information across sites with a light enough tail to keep (34) finite; a useful secondary check.
- **Fréchet** margins are the native max-stable scale but a poor scoring scale — the heavy tail lets a single extreme dominate even at $p = 0.5$; avoid unless scoring a max-stable model directly.

Lemma 4 (PIT degeneracy)**.** *If $\hat{F}_{s,t}$ is taken as the empirical CDF of the same m posterior predictive draws that are then scored, the PIT-transformed draws are the ranks $\{1/m, \ldots, 1\}$ — a deterministic set independent of the dependence structure, so the score cannot discriminate any two methods.*

The fix is to define $\hat{F}_{s,t}$ from a source disjoint from the scored sample: either (a) split the predictive draws, using a held-back subsample to estimate $\hat{F}_{s,t}$ and the remainder to score, or (b) use a parametric tail surrogate — fit a GPD or skew-t to the upper-tail draws and use that smooth CDF for both the observation and the scored draws. Compute $\hat{F}_{s,t}$ once per (s, t, method) and cache it.

10.6 Design C (recommended): combined holdout + score triple

Hold out a set of test sites *and* a terminal time block at the remaining sites, PIT-transform to uniform margins, and report a *triple*:

1. $\overline{\text{CRPS}}$ on (s^*, t), $t \le T - T_h$ — the marginal-calibration control (Proposition 11);
2. per-horizon energy score on training sites over $t > T - T_h$ — the temporal-coupling test;
3. per-horizon variogram score $\text{VS}_{0.5}$ on $(s^*, t > T - T_h)$ — space-and-time dependence.

The attribution that has power: if $\overline{\text{CRPS}}$ is equal across methods but $\text{VS}_{0.5}$ differs, the difference is *purely dependence*. Because the level is white (Corollary 6), the only score in the triple that can reward the true (τ, w) structure is one applied to $|Y|^p$ or to volatility-sensitive functionals; a VS_p with $p \approx 1$ on the levels, or any score on Y^2, is the route to the τ-AR(2) signal (Corollary 4).

10.7 Detectability and reporting

Let $\Delta = S_{\text{no-AR}} - S_{\text{AR}}$ be the paired score difference, averaged over N holdout blocks. Its standard error scales as $\text{SE}(\Delta) \propto N^{-1/2}$, and the detectability index is $\Delta/\text{SE}(\Delta)$. The expected Δ on the CRPS is bounded by the ceiling $[1 - \sqrt{1 - R(h)}] \cdot \text{CRPS}_{\text{no-AR}}$ of Proposition 6. Practical rule: from a pilot estimate of $\text{SNR}_{\text{temporal}}$ and $\kappa_z(h)$, compute the ceiling, then choose N (via rolling-origin blocks) so that $\text{SE}(\Delta)$ sits a few times below it. Report paired differences with bootstrap intervals, per horizon, and a score-per-CPU-hour axis — methods 7 and 8 are the more expensive.

11 Conclusion and Future Work

- The AR(2) standardisation makes the prior time-t marginal of every latent ϕ-invariant (Propositions 1, 3); the spatial-only Brier integrates the kernel against that marginal (Proposition 2), so the temporal coupling marginalises out.
- Even a frozen true ϕ cannot dominate: the posterior-pooling benefit is $O(q_t^{-2})$ and never touches the ϕ-invariant Channel-D floor (Proposition 4, Corollary 3).

- AR(2) reaches the *level* of Y through one channel only, $\mathrm{Cov}(\widetilde{Y}_t, \widetilde{Y}_{t+h}) = \lambda^2 \gamma_z(h)$ (Proposition 5); the τ-AR(2) is pure stochastic volatility (Corollary 4). Because the simulation truth sets $\phi_z = (0,0)$, the level is exactly white in time and no level-based score can reward AR(2) for any holdout design (Corollary 6); the achievable gain is capped by $\mathrm{SNR}_{\mathrm{temporal}}$ with the closed-form CRPS ceiling $1 - \sqrt{1 - R(h)}$ (Proposition 6).
- A freely mixing ϕ helps the *other* parameters converge by softening a stiff conditional target and acting as a parameter-expansion coordinate (Proposition 7, Remarks 5–6).
- A misspecified ϕ still reduces posterior variance — the SD plot is robust to a wrong ϕ (Proposition 8) — but pays in bias; collapse to i.i.d. occurs only as $\phi \to 0$ (Proposition 9).
- The observed collapse $\hat{\phi} \to 0$ is errors-in-variables attenuation (Lemma 3): ϕ-information is monotone increasing in q_t (Proposition 10), and the simulation truth places the signal in the lowest-q_t latents (Eq. (30)).

Future work, in order of priority:

1. **Compute the ceiling before re-scoring.** From the true parameters of settings 9–10, evaluate $\mathrm{SNR}_{\mathrm{temporal}} = \lambda^2 \,\mathrm{Var}(z)/\,\mathrm{Var}(\widetilde{Y})$ and $\kappa_z(h)$ (Definition 4); overlay $1 - \sqrt{1 - R(h)}$ on any per-horizon score curve as the theoretical bound. If the ceiling never exceeds $\mathrm{SE}(\Delta)$, that is itself the publishable result.
2. **Run the four-layer ACF diagnostic** (Appendix C) to confirm where the AR(2) signal is lost — source latent, membership reindexing, λ-gating, or the i.i.d. spatial noise.
3. **Implement Design C** in `scores.R` and `run-settings.R`: a terminal-block holdout, the predictive loop of Remark 9, PIT standardisation (Section 10.5), and `scoringRules::{es,vs,crps}_sample`.
4. **Score volatility, not just level.** Since the true temporal signal of settings 9–10 lives in τ (Corollary 4), apply $\mathrm{VS}_{p\approx 1}$ or a score on Y^2 — the only route by which the τ-AR(2) structure becomes visible.
5. **Confirm the attenuation:** compute the lag-1 autocorrelation of the posterior-mean $\tau^\star$ series and divide by $\gamma_1 = 0.5$ to recover $a = q/(q+1)$; profile $\log L(\phi)$ on a grid to check for flatness near 0.
6. **Raise $\mathcal{I}(\phi)$:** more replicate series (larger K or T), a non-centred parameterisation of $\tau^\star$ to break the Gibbs fixed point, a ϕ-prior not concentrated at 0, or marginalising the latent series so ϕ is informed by $L(\phi)$ directly.
7. Run AR-order selection over $p \in \{0, 1, 2\}$ via DIC/WAIC so any observed advantage is not an artefact of a matched truth.

A Reference R implementation

The standardised AR(2) simulator in `code/00_core/auxfunctions_ar2.R`.

```
simulate_ar2_standard <- function(nt, n, phi1, phi2,
                                  tol = 1e-06, param_name = NULL) {
  # ... stationarity check ...
  denom  <- 1 - phi2
  gamma0 <- 1
  gamma1 <- phi1 * gamma0 / denom
  gamma2 <- phi1 * gamma1 + phi2 * gamma0
  sigma2 <- gamma0 - phi1 * gamma1 - phi2 * gamma2
```

```
  innov.sd <- sqrt(sigma2)

  ar2 <- matrix(0, nrow = n, ncol = nt)
  init.states <- draw_ar2_initial(n, gamma0, gamma1)
  ar2[, 1] <- init.states[, 1]
  ar2[, 2] <- init.states[, 2]

  for (t in 3:nt) {
    ar2[, t] <- phi1 * ar2[, t - 1] +
                phi2 * ar2[, t - 2] +
                rnorm(n, 0, innov.sd)
  }
  ar2
}
```

The construction matches Definition 1: the marginal variance is fixed at $\gamma_0 = 1$, the innovation variance is derived, and initial states are drawn from the stationary bivariate Gaussian via `draw_ar2_initial()`, removing the need for a burn-in.

B Verification by reverse simulation

To confirm the marginal invariance empirically we simulate $n = 1000$ AR(2) paths under $(\phi_1, \phi_2) = (0.6, -0.3)$ via manual recursion and via `stats::filter(..., method = "recursive")`, fit `arima(order = c(2, 0, 0))` to each, and run a residual diagnostic. Both fits recover the true coefficients within one standard error, and the Ljung–Box and Shapiro–Wilk tests on the residuals are non-significant ($p > 0.4$). Manual recursion and filtering produce statistically equivalent AR(2) series.

C Four-layer ACF attenuation diagnostic

To localise where the AR(2) signal is lost on the path from the latent z to the observable Y, decompose a method-8 posterior predictive trajectory at a held-out site s^* into four layers and compare their empirical autocorrelations:

L1 $z_{k^*,t}$ — the source latent at a fixed knot; should match the theoretical AR(2) ACF implied by $\hat{\phi}_z$.

L2 $z_{g(s^*,t),t}$ — after membership reindexing; attenuation here measures knot-wandering loss.

L3 $\mu_{s^*,t} = x_{s^*,t}^\top \beta + \lambda\, z_{g(s^*,t),t}$ — after λ-gating.

L4 $Y_t(s^*)$ — the full predictive draw, after the i.i.d.-in-t spatial noise; this is where most of the signal is expected to die.

```
# Four-layer ACF attenuation diagnostic for one method-8 fit.
draws <- sample(seq_along(fit.1$lambda), 50)
k_star <- 1
for (d in draws) {
  beta_d    <- fit.1$beta[d, ]
  lambda_d  <- fit.1$lambda[d]
  z_d       <- fit.1$z[, , d]
  knots_d   <- fit.1$knots[, , , d]

  L1 <- z_d[k_star, ]
  L2 <- sapply(seq_len(nt), function(t)
```

```
        z_d[mem(matrix(s.p[sp, ], 1, 2),
                matrix(knots_d[, , t], nknots, 2)), t])
  L3 <- as.numeric(x.p[sp, , ] %*% beta_d) + lambda_d * L2
  L4 <- L3 + rnorm(nt, 0, sigma_g_t)        # iid spatial noise

  store(acf(L1, plot = FALSE)$acf[2:4],
        acf(L2, plot = FALSE)$acf[2:4],
        acf(L3, plot = FALSE)$acf[2:4],
        acf(L4, plot = FALSE)$acf[2:4])
}
# Average across draws; compare L1 to ARMAacf(ar = colMeans(fit.1$phi.z)).
```

If L1 matches the theoretical ACF but L4 is essentially white, the signal is lost to the λ-gating and the spatial noise — in agreement with Proposition 5 — and no level-based score can recover it. Use posterior *parameter* draws but the stored latent fields z, τ, w within each draw; resampling the latents from their AR(2) prior would report the prior ACF rather than the fitted one.

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